

TUSHAR KHARI

Data Analyst / Machine Learning Engineer

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Portfolio



Education

Hochschule Aalen, Germany	2025 – Present
M.Sc. in Machine Learning and Data Analytics	GPA: 1.9 (German Scale)
Panjab University, India	2020 – 2024
B.Eng. in Information Technology	CGPA: 8.47 / 10

Work Experience

Software Engineer	Jan 2024 – Dec 2024
ICS Group	Chandigarh, India

- Deployed containerized FastAPI services using Docker to serve deep learning models with < 120 ms response latency.
- Built and fine-tuned PyTorch neural network architectures, boosting classification accuracy by 17%.
- Developed an internal LLM retrieval system using LangChain to automate unstructured document processing.
- Optimized automated Python and SQL ETL pipelines, reducing training data prep time by 30%.
- Authored automated pytest suites (85% test coverage) and CI/CD pipelines to ensure release stability.

Key Projects

Predictive Maintenance System Python, Scikit-learn, FastAPI, Streamlit	[Live App]
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- Trained and evaluated Random Forest and XGBoost models on the AI4I dataset using ROC-AUC and F1 scores.
- Tuned decision thresholds to 0.75, improving critical equipment failure recall by 15%.

Modern Data Warehouse & Analytics Pipeline PostgreSQL, ETL, Data Modeling	[GitHub]
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- Built an end-to-end data warehouse implementing Medallion Architecture across ERP and CRM sources.
- Structured dimensional star schemas and optimized analytical SQL queries for business reporting.

Multimodal Sensitive Data Detection PyTorch, YOLO, BLIP-2, Streamlit	[GitHub]
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- Implemented a privacy pipeline combining YOLO object detection with OCR and zero-shot NLP classification.
- Integrated BLIP-2 for context-aware visual privacy analysis within an interactive Streamlit UI.

Technical Skills

Languages & Frameworks: Python, SQL, Java, PyTorch, Scikit-learn, LangChain, FastAPI

Data Engineering & BI: PostgreSQL, Databricks, PySpark, Power BI, ETL Pipelines

Tools & Cloud: Git, Docker, Azure, Linux, pytest, CI/CD, Streamlit

Languages: English (Fluent), German (B1, progressing to B2)